# Navyug Vidyabhavan Trust

## C. K. Pithawala College of Engineering & Technology

Computer Engineering Department

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# AI Based Internship Recommendation system

**Subject:**
**Data Mining**
(3160714)
BE III, Sem VI

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# Outline

- Problem Statement
- Abstract / Overview
- Dataset and Research paper links
- Preprocessing techniques
- Association rule mining
- Supervised learning algorithms
- Unsupervised learning algorithms
- Text mining techniques
- Web mining techniques
- Results-comparison-Graphs
- Conclusion
- References

# Problem Statement

Students often struggle to find suitable internships due to the large number of opportunities available across different platforms and the lack of proper guidance based on their skills and interests. Most existing systems rely on manual searching or simple keyword matching, which can be time-consuming and inefficient. There is a need for an intelligent system that can automatically analyze internship data and recommend the most relevant opportunities to students. This project proposes an AI-based Internship Recommendation System that uses data mining techniques to process internship details such as required skills, role, location, duration, and provide personalized and fair recommendations to users.

# Abstract/Overview

- The system recommends suitable internships to students based on their skills, experience, and preferences.
- Traditional platforms use keyword matching, which often gives less accurate and inefficient results.
- Data mining techniques such as preprocessing, classification, clustering, and text mining are used to analyze internship data.
- The system identifies patterns between skills and internship requirements to provide accurate recommendations.
- The output is a ranked list of relevant internships, ensuring fair and unbiased recommendations.

# Dataset and Research paper links

1. **Dataset :**

   https://drive.google.com/file/d/1JXFYD1e8qEoYF0KDMOQrfqpcG1GadYSG/view?usp=sharing

2. **Research paper :**

   https://dl.acm.org/doi/pdf/10.1145/3700297.3700386

   https://www.emerald.com/jrit/article-pdf/17/2/352/9542864/jrit-01-2024-0016.pdf

**Dataset and research references used to support preprocessing, clustering, and classification techniques in the proposed system.**

# Preprocessing techniques

## Techniques:

1. **Data Cleaning:** Handling missing values, removing duplicates, Example: ₹10,000/month → 10000
2. **Data Transformation:** Converted categorical values, Example: Online = 1, Offline = 0
3. **Data Integration:** Merging internship datasets from different sources
4. **Data Reduction & Feature Selection:** Selected important features (skills, role, stipend, location)
5. **Feature Extraction:** Converting text (skills, job description) into numerical format

## Language/Tools used:

Python (Pandas, NumPy, Matplotlib, Seaborn), Google colab

# Association rule mining

**Technique Detail (with Numerical Example):**

Apriori is used to find relationships between skills and internship roles.

**Example:**

- [Python, ML, Data Science Intern]
- [Python, SQL, Data Analyst Intern]

**Rule generated:**

{Python} → {Data Science Intern}

**Support:** 0.5 (appears in 50% data)

**Confidence:** 0.67 (67% of Python internships are Data Science)

**Language/Tools used:** Python, Pandas, mlxtend, google colab

# Supervised learning algorithms

**Classification (Techniques from Syllabus):**

- Algorithms: KNN, Naive Bayes, Decision Tree, Neural Network
- Example (dataset):
  - Input: Stipend = 10000, Duration = 3 → Output: Data Science Intern
- KNN: classifies based on nearest data points
- Decision Tree: splits data using conditions

**Language / Tools Used:**

- Python, Scikit-learn, Pandas, Colab

**Technique from Latest Research Paper:**

- Random Forest Classifier
- Example: Combines multiple decision trees for better accuracy

**Language / Tools Used:**

- Python, Scikit-learn, Colab

# Unsupervised learning algorithms

**Techniques from Syllabus:**

- **K-Means (Distance-based):**
  - Example:
    - (10000, 3) and (12000, 4) → same cluster
    - (5000, 2) → different cluster
- **Hierarchical (Agglomerative):**
  - Starts with individual points and merges similar internships step-by-step

**Technique from Latest Research Paper:**

- **DBSCAN (Density-based):**
  - Example:
    - Dense points → cluster
    - Isolated point → noise (-1)

**Algorithm Detail (Dataset Use):**

- Input: Stipend, Duration, Openings
- Output: Grouped internships into clusters (e.g., high stipend, low stipend)

**Language / Tools Used:**

- Python
- Scikit-learn
- Pandas
- Google Colab

# Text mining techniques

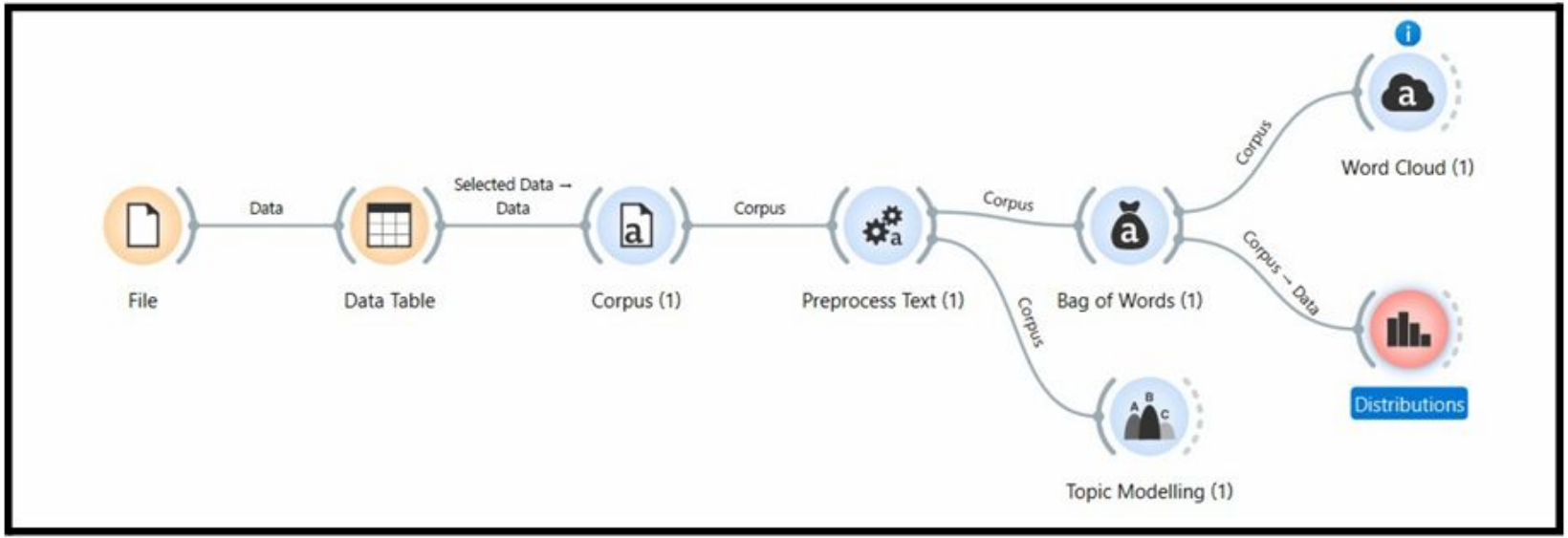
**Technique Detail (with Numerical Example)**

- **Technique Used:** TF-IDF (Text Mining)
- Converts text (skills) into numerical values
- **Example (from dataset):**
  - **Input:** “Python, Machine Learning”
  - **Output:** [0.72, 0.45] (importance scores)
- Helps identify important skills for internships
- Used for better analysis and recommendation

**Language / Tools Used:**

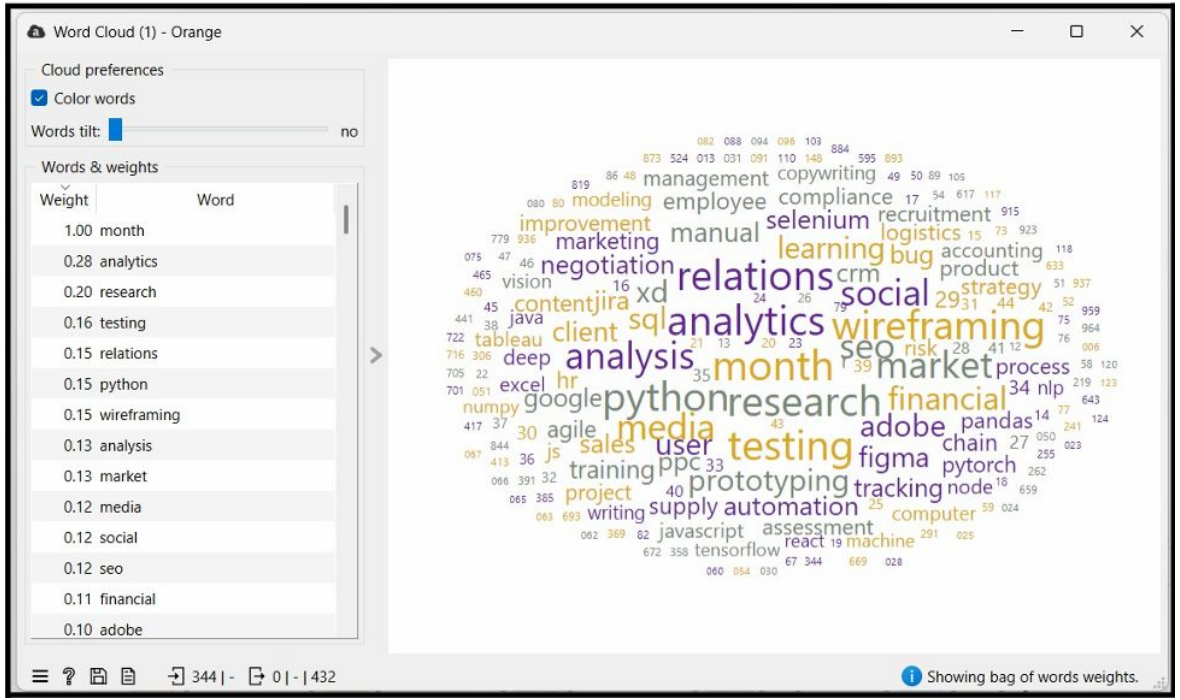
- Python
- Scikit-learn
- Pandas
- Google Colab
- Orange

# Text mining techniques Implementation in Orange



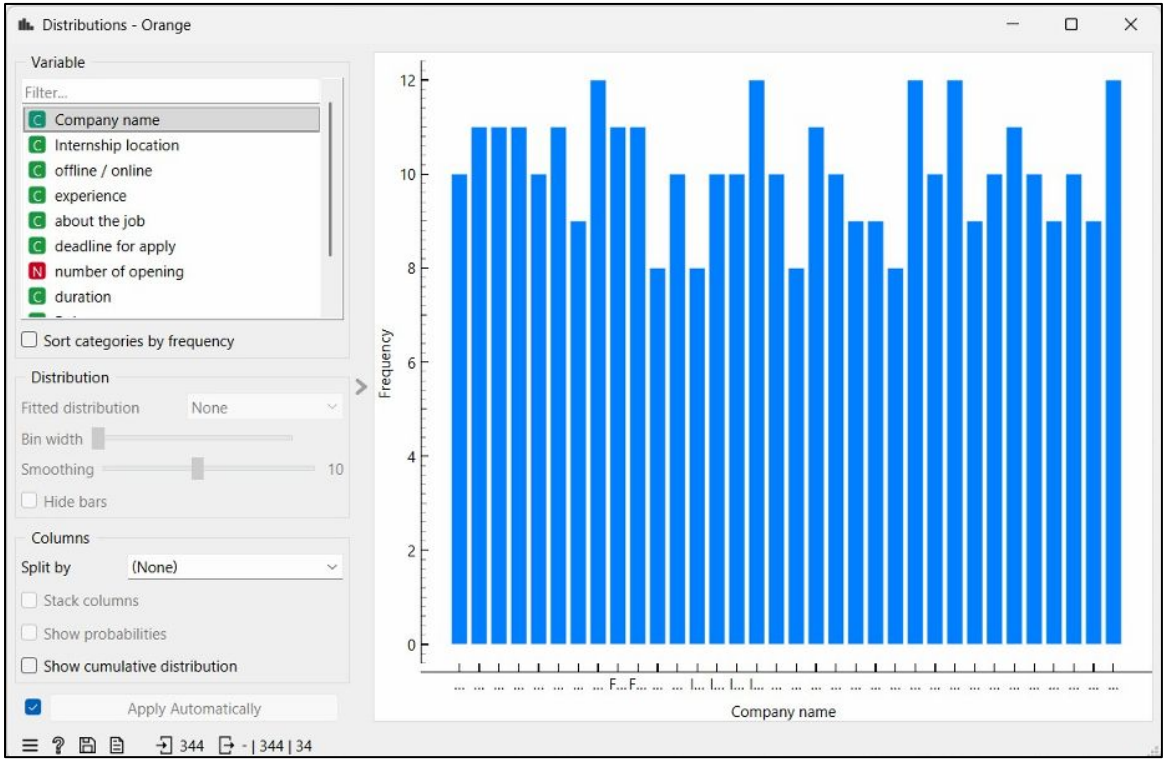

**Text Mining Workflow**

This image shows an Orange workflow for text mining, including preprocessing, bag of words, word cloud, and topic modeling to extract insights from textual data.




**Word Cloud Output**

This image displays a word cloud visualization highlighting the most frequent and important terms like “analysis”, “python”, and “research” from the dataset.




**Company Distribution Chart**

This image shows a frequency distribution of different company names using a bar chart in the Orange tool. Each bar represents how often a company appears in the dataset.

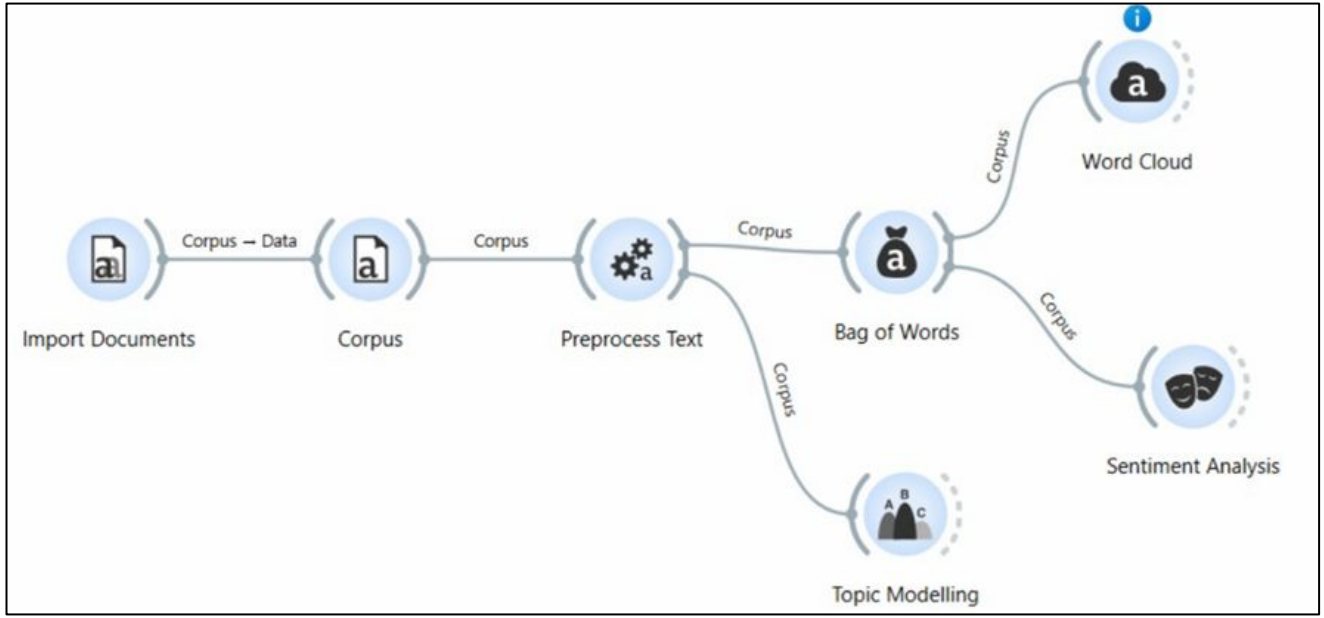
# Web mining techniques

**Technique Detail (with Numerical Example)**

- **Technique Used:** Web Mining (API-based Data Extraction)
- Extracts real-time job data from websites
- **Example (from dataset):**
  - **Input (API data):** Company = Google, Role = Data Analyst, Skills = Python, SQL
  - **Output:** Stored as structured data in table format
- Helps identify top companies and in-demand skills

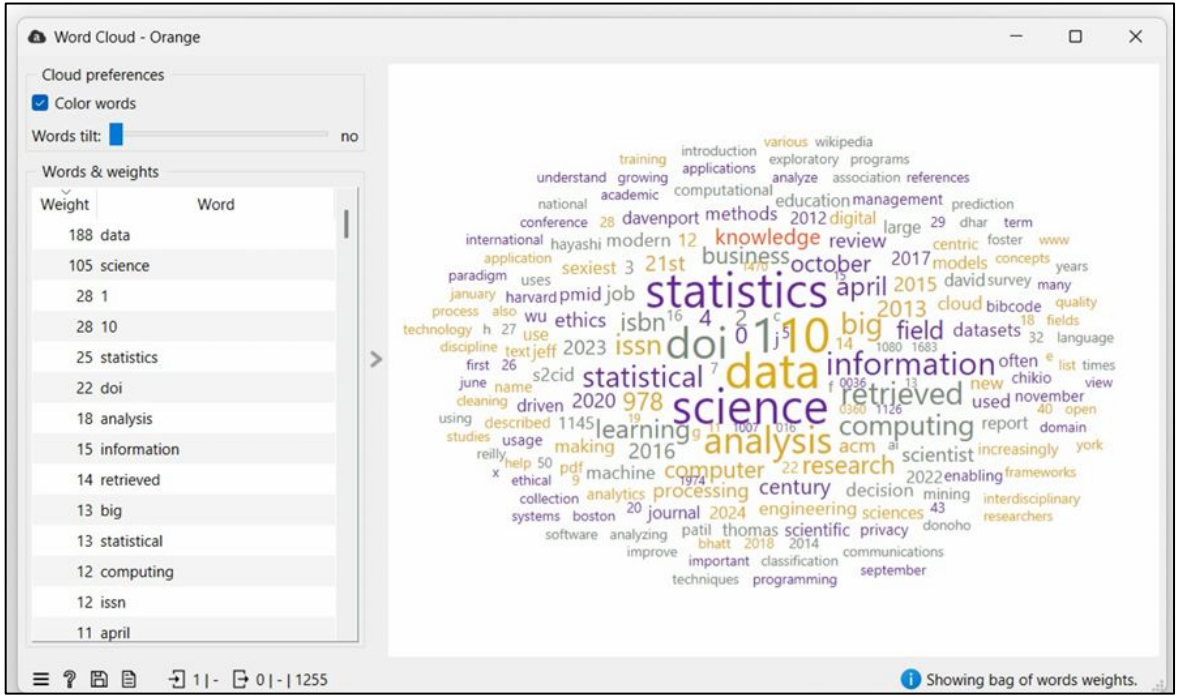
**Language / Tools Used:**

- Python
- Requests (API fetching)
- Pandas (data handling)
- Google Colab
- Orange



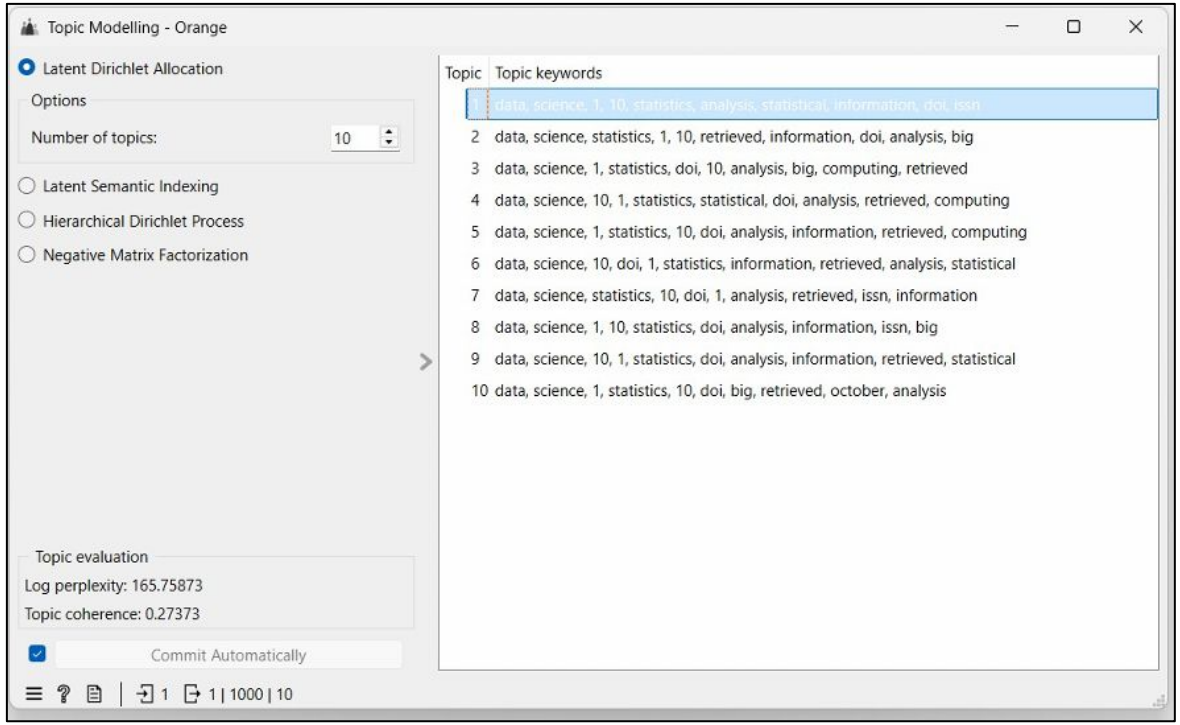

**Text Analysis Workflow**

This image shows a text mining pipeline in Orange, starting from document import to preprocessing and analysis. It includes steps like bag of words, word cloud, sentiment analysis, and topic modeling.




**Word Cloud**

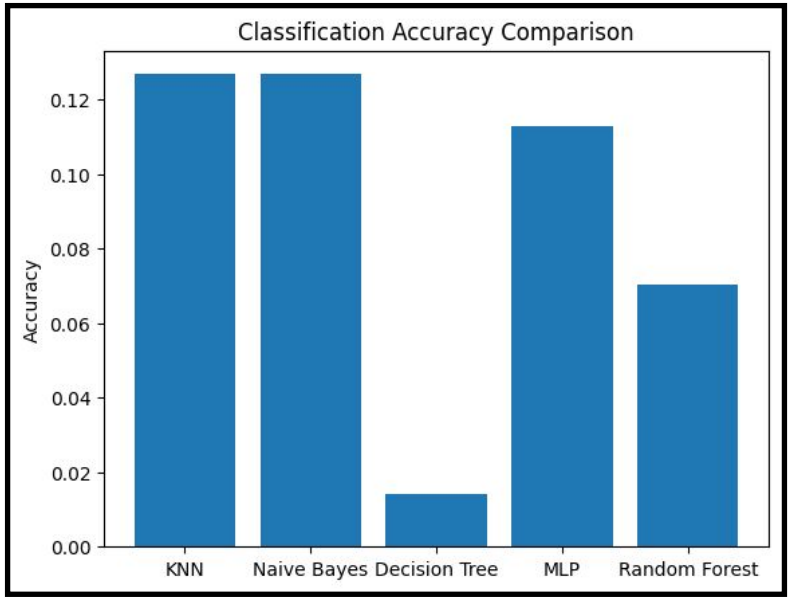
This image shows a word cloud visualization generated using Orange, highlighting the most frequent and important words in the dataset. Larger words like "data" and "science" indicate higher importance.

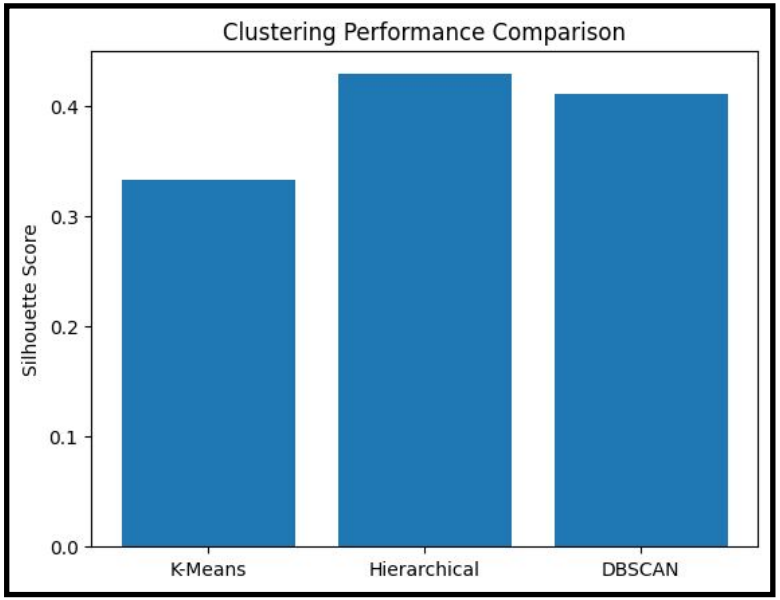



This image represents topic modeling using Latent Dirichlet Allocation (LDA) in Orange. It displays groups of keywords forming different topics extracted from the dataset.

**Topic Modeling**

# Results-comparison-Tables-Graphs

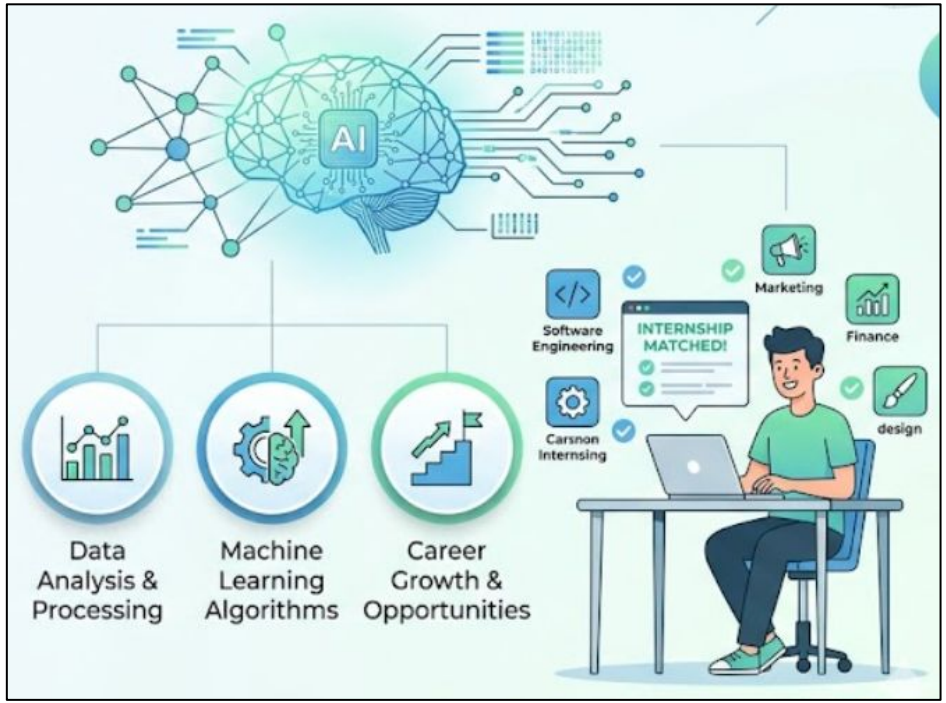



**The comparison shows that KNN and Naive Bayes perform best for classification, while Hierarchical clustering provides better cluster separation.**

# Conclusion

This project demonstrates the use of data mining and machine learning techniques to build an AI-Based Internship Recommendation System. It provides personalized and data-driven internship recommendations based on skills and preferences. The system helps reduce manual effort and improves decision-making for students, showing the impact of AI in solving real-world problems.



# References:


1. Smart India Hackathon-2025 Problem Statements, Available Online at: https://sih.gov.in/sih2025PS
2. Dataset used: https://drive.google.com/file/d/1JXFYD1e8qEoYF0KDMOQrfqpcG1GadYSG/view?usp=sharing
3. Research on Intelligent Employment Recommendation for College Students Based on Improved DPC: https://dl.acm.org/doi/pdf/10.1145/3700297.3700386
4. Navigating techniques in job recommender systems on internship profile matching: a systematic review: https://www.emerald.com/jrit/article-pdf/17/2/352/9542864/jrit-01-2024-0016.pdf
5. Code Implementation: https://colab.research.google.com/drive/1IXmCBaId0idPByE8hvua5SRduwc4dKQg?usp=sharing
6. Report: https://docs.google.com/document/d/13K70neqVrmACpfqiFWYTFxHePT4gDMryGmlGKSF53zA/edit?usp=sharing